```python
#Train reservation seat
seats = [0] * 20
while True:
    print("\n1.Book Seat\n2.Cancel Seat\n3.Show Seats\n4.Booking Percentage\n5.Exit")
    choice = int(input("Enter choice: "))

    if choice == 1:
        seat_no = int(input("Enter seat number (1-20): ")) - 1
        if 0 <= seat_no < 20:
            if seats[seat_no] == 0:
                seats[seat_no] = 1
                print("Seat booked successfully!")
            else:
                print("Seat already booked.")
        else:
            print("Invalid seat number.")

    elif choice == 2:
```

```
        seat_no = int(input("Enter seat number (1-20): ")) - 1
        if 0 <= seat_no < 20:
            if seats[seat_no] == 1:
                seats[seat_no] = 0
                print("Reservation cancelled.")
            else:
                print("Seat was not booked.")
        else:
            print("Invalid seat number.")

    elif choice == 3:
        print("Seat Map (0=Available, 1=Booked):")
```

```
        for i, s in enumerate(seats):
            print(f"Seat {i + 1}:{s}", end=" | ")
        print()

    elif choice == 4:
        booked_count = seats.count(1)
        percentage = (booked_count / 20) * 100
        print(f"Booking Percentage: {percentage}%")

    elif choice == 5:
        break
    else:
        print("Invalid choice.")
```

## Output

```
1.Book Seat
2.Cancel Seat
3.Show Seats
4.Booking Percentage
5.Exit
Enter choice:
```

```python
#youtube playlist analyser
durations = [12, 4, 8, 15, 3, 20]
# Calculations
total_time = sum(durations)
longest_video = max(durations)
avg_duration = total_time / len(durations)

shorter_than_5 = []
for d in durations:
    if d < 5:
        shorter_than_5.append(d)
# Results
print(f"Total Watch Time: {total_time} minutes")
print(f"Longest Video: {longest_video} minutes")
print(f"Average Video Duration: {avg_duration:.2f} minutes")
print(f"Videos shorter than 5 minutes: {shorter_than_5}")
```

Output

```
Total Watch Time: 62 minutes
Longest Video: 20 minutes
Average Video Duration: 10.33 minutes
Videos shorter than 5 minutes: [4, 3]

=== Code Execution Successful ===
```

```python
#Social media hastag hacker
hashtags = ["#fun", "#python", "#fun", "#code", "#python"]

# Frequency distribution using lists
unique_tags = []
frequencies = []

for tag in hashtags:
    if tag not in unique_tags:
        unique_tags.append(tag)
        frequencies.append(hashtags.count(tag))

# Find the most repeated hashtag
max_count = max(frequencies)
most_repeated = unique_tags[frequencies.index(max_count)]

# Results
print(f"Unique hashtags: {unique_tags}")
print(f"Most repeated hashtag: {most_repeated}")
```

```python
print("Frequency of each hashtag:")
for i in range(len(unique_tags)):
    print(f" {unique_tags[i]}: {frequencies[i]}")
print(f"List after removing duplicates: {unique_tags}")
```

Output

```
Unique hashtags: ['#fun', '#python', '#code']
Most repeated hashtag: #fun
Frequency of each hashtag:
 #fun: 2
 #python: 2
 #code: 1
List after removing duplicates: ['#fun', '#python', '#code']

=== Code Execution Successful ===
```

```python
#Expense splitter
expenses = [1200, 850, 1500, 950]
friends = ["Friend A", "Friend B", "Friend C", "Friend D"]
total_expense = sum(expenses)
num_friends = len(expenses)
avg_share = total_expense / num_friends
# Highest payer tracking
max_expense = max(expenses)
highest_payer = friends[expenses.index(max_expense)]
# Results
print(f"Total Expense: {total_expense}")
print(f"Average Share (Amount each should contribute equally): {avg_share}")
print(f"Person who paid highest: {highest_payer} ({max_expense})")
```

Output

```
Total Expense: 4500
Average Share (Amount each should contribute equally): 1125.0
Person who paid highest: Friend C (1500)

=== Code Execution Successful ===
```

```python
#pattern comparision
input_list = [1, 1, 1, 2, 2, 3, 3, 3]
output_list = []
if input_list:
    current_element = input_list[0]
    count = 1
    for i in range(1, len(input_list)):
        if input_list[i] == current_element:
            count += 1
        else:
            output_list.append((current_element, count))
            current_element = input_list[i]
            count = 1
    output_list.append((current_element, count))  # Append final group
print(f"Input : {input_list}")
print(f"Output: {output_list}")
```

## Output

```
Input : [1, 1, 1, 2, 2, 3, 3, 3]
Output: [(1, 3), (2, 2), (3, 3)]

=== Code Execution Successful ===
```

```python
#Mini shopping cart system
cart = []
while True:
    print("\n1.Add item\n2.Remove item\n3.Search item\n4.Calculate total bill\n5.Display duplicates\n6.Exit")
    choice = int(input("Enter choice: "))
    if choice == 1:
        item = input("Enter item name: ").lower()
        price = float(input("Enter item price: "))
        cart.append([item, price])
        print(f"{item} added to cart.")
    elif choice == 2:
        item = input("Enter item to remove: ").lower()
        removed = False
        for pair in cart:
            if pair[0] == item:
                cart.remove(pair)
                print(f"{item} removed from cart.")
                removed = True
```

```
            break
        if not removed:
            print("Item not found in cart.")
    elif choice == 3:
        item = input("Enter item to search: ").lower()
        found = False
        for pair in cart:
            if pair[0] == item:
                print(f"Found! {item} costs {pair[1]}")
                found = True
                break
        if not found:
            print("Item not found.")
    elif choice == 4:
        total = sum(pair[1] for pair in cart)
        print(f"Total Bill: {total}")
    elif choice == 5:
        # Track items to find accidental duplicates
        seen = []
```

```
        duplicates = []
        for pair in cart:
            if pair[0] in seen and pair[0] not in duplicates:
                duplicates.append(pair[0])
            seen.append(pair[0])
        print(f"Duplicate items added accidentally: {duplicates}")
    elif choice == 6:
        break
```

## Output

```
1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 1
Enter item name: orange
Enter item price: 100
orange added to cart.
```

```
1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 2
Enter item to remove: 3
Item not found in cart.
```

```
1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 3
Enter item to search: orange
Found! orange costs 100.0
```

```
1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 4
Total Bill: 100.0
```

```
1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 5
Duplicate items added accidentally: []

1.Add item
2.Remove item
3.Search item
4.Calculate total bill
5.Display duplicates
6.Exit
Enter choice: 6

=== Code Execution Successful ===
```